Convergence of Markov Chain Monte Carlo Algorithms

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1 Introduction

The goal is to address when an MCMC algorithm will produce a representative sample from the invariant distribution F. Slightly more formally, if $\text{dist}(P^n, F)$ denotes the distance between the distribution of the X_n and F the interest is in the following three questions:

1. When does
$$\lim_{n\to\infty} \text{dist}(P^n, F) = 0?$$
2. Given ϵ, for what n does
$$\text{dist}(P^n, F) \le \epsilon?$$
3. How fast does the convergence in item 1 occur?

The focus of this module will be on the first question. The second and third questions are closely related and will be considered in detail later.

2 Total Variation

Several metrics for measuring distance will be encountered throughout, but the most common is the total variation norm, denoted $\|\cdot\|$. Let ν_1 and ν_2 be probability measures on $(\mathsf{X}, \mathcal{B}(\mathsf{X}))$ then

$$\|\nu_1 - \nu_2\| = \sup_{A\in\mathcal{B}(\mathsf{X})} |\nu_1(A) - \nu_2(A)|.$$

In the context of MCMC, the interest will lie in the behavior of, as $n \to \infty$,

$$\|P^n(x,\cdot) - F(\cdot)\|.$$

Notice that if

$$\lim_{n\to\infty} \|P^n(x,\cdot) - F(\cdot)\| = 0,$$

then this also implies convergence in distribution. Hence the MCMC simulation will eventually produce a representative sample from F.

An equivalent formulation of total variation will be useful, but a definition is needed first. Define

$$\nu_1 f = \int_{\mathsf{X}} f(x)\nu_1(\mathrm{d}x)$$

and, in particular,

$$P^n f(x) = \int_{\mathsf{X}} f(y) P^n(x, \mathrm{d}y) = E\left[f(X_{n+1}) \mid X_n = x\right].$$

Say that Markov kernels act to the right on functions.

The following is equivalent to the definition of total variation given above:

$$\|\nu_1 - \nu_2\| = \sup_{f:\mathsf{X}\to[0,1]} |\nu_1 f - \nu_2 f|.$$

Proposition 2.1. *Let ν_1 and ν_2 be probability measures, P a Markov kernel and F its invariant distribution. Then the following hold:*

1. *$\|P^n(x, \cdot) - F(\cdot)\|$ is non-increasing in n and*

2. *$\|\nu_1 P - \nu_2 P\| \leq \|\nu_1 - \nu_2\|$.*

Proof. Only the first claim is considered here; the proof of the second claim is left as an exercise. Notice that $f(y) = P(y, A) \in [0, 1]$ so that

$$\begin{aligned} |P^{n+1}(x, A) - F(A)| &= \left| \int_{\mathsf{X}} P^n(x, \mathrm{d}y) P(y, A) - \int_{\mathsf{X}} F(\mathrm{d}y) P(y, A) \right| \\ &= \left| \int_{\mathsf{X}} P^n(x, \mathrm{d}y) f(y) - \int_{\mathsf{X}} F(\mathrm{d}y) f(y) \right| \\ &\leq \|P^n(x, \cdot) - F(\cdot)\|. \end{aligned}$$

□

3 Aperiodic Ergodic Theorem

To this point the emphasis has been on F-irreducibility and the invariance of F for the kernel P. This is often sufficient in MCMC settings, as will be discussed below, but in general more is required to produce a representative sample.

Definition 3.1. The kernel P is *aperiodic* if there do not exist $d \geq 2$ and disjoint sets $A_1, \ldots, A_d \subseteq \mathsf{X}$ such that each $F(A_i) > 0$, $P(x, A_{i+1}) = 1$ for all $x \in A_i$, $1 \leq i \leq d-1$ and $P(x, A_1) = 1$ for all $x \in A_d$. If $N = (A_1 \cup \cdots \cup A_d)^c$, then $F(N) = 0$. Otherwise, the kernel is *periodic*.

Theorem 3.1. *If P has invariant distribution F, is aperiodic, and F-irreducible, then for F-almost all $x \in \mathsf{X}$*

$$\lim_{n\to\infty} \|P^n(x,\cdot) - F(\cdot)\| = 0.$$

A proof will be discussed later. Periodicity in practically relevant MCMC is rare.

Proposition 3.2. *Suppose P is F-irreducible and that $A \in \mathcal{B}(\mathsf{X})$ is such that $F(A) > 0$. Let $x \in A$ and assume that $P(x, B) > 0$ for all $B \subseteq A$ such that $x \in B$ and $F(B) > 0$. Then P is aperiodic.*

The proof of the proposition is left as an exercise.

Corollary 3.1. *If there exists S such that $F(S) > 0$ and $P(x, \{x\}) > 0$ for all $x \in S$, then P is aperiodic.*

The following result was first conveyed to me by Charlie Geyer. I don't believe it has been published.

Proposition 3.3. *An F-irreducible Gibbs sampler is aperiodic.*

Proof. Consider a random scan Gibbs sampler (deterministic scan is left as an exercise) and suppose by way of contradiction that it is periodic with period d. Then there exists $A_1, \ldots, A_d \subseteq \mathsf{X}$ such that each $F(A_i) > 0$, $P(x, A_{i+1}) = 1$ for all $x \in A_i$, $1 \le i \le d-1$ and $P(x, A_1) = 1$ for all $x \in A_d$. In P_{RG}, one component is chosen at a time to update. Let I_n be this random choice and let $h_{I_n}(X_n)$ be the part of X_n which is not updated. Then X_n and X_{n+1} are conditionally independent given $h_{I_n}(X_n)$. Therefore,

$$\Pr(X_{n+1} \in A_k \mid X_n \in A_k, h_{I_n}(X_n)) = \Pr(X_{n+1} \in A_k \mid h_{I_n}(X_n)). \tag{1}$$

Also, by periodicity, $\Pr(X_{n+1} \in A_k \mid X_n \in A_k) = 0$. But

$$\Pr(X_{n+1} \in A_k \mid X_n) = E\left[\Pr(X_{n+1} \in A_k \mid X_n \in A_k, h_{I_n}(X_n)) \mid X_n \in A_k\right]$$

and hence the expression in (1) is F-almost surely 0. In turn, this implies $\Pr(X_{n+1} \in A_k) = 0$, which is a contradiction. $\square$

Irreducible Gibbs samplers, Metropolis-Hastings, and conditional Metropolis-Hastings chains will typically be aperiodic and hence the results of Theorem 3.1 obtain. However, there is an annoying null set where the convergence may not happen.

Example 3.1. This example is taken from existing work [4]. Suppose $\mathsf{X} = [0, 1]$. If $x = 1/m$ for positive integers m, let

$$P(x, \cdot) = x^2 \text{Uniform}[0, 1] + (1 - x^2)\delta_{\frac{1}{m+1}}(\cdot)$$

otherwise

$$P(x, \cdot) = \text{Uniform}[0, 1].$$

Then $F = \text{Uniform}[0, 1]$ and $FP = F$, P is F-irreducible, and aperiodic. But, if $X_0 = \frac{1}{m}$ for any $m \geq 2$, then

$$\Pr\left(\left\{X_n = \frac{1}{m+n} \text{ for all } n\right\}\right) = \prod_{j=m}^{\infty}\left(1 - \frac{1}{j^2}\right) > 0.$$

Hence $\|P^n(1/m, \cdot) - F(\cdot)\|$ does not converge to 0.

Another property is required to do away with the null set where convergence is problematic. This si addressed in the next section.

4 Harris Recurrence

Let $A \in \mathcal{B}(\mathsf{X})$ and define

$$\tau_A = \inf\{n \geq 1 \mid X_n \in A\}$$

so that τ_A is the first return time to A. Note that $\tau_A = \infty$ if $X_n \notin A$ for all $n \geq 1$.

If $FP = F$ and P is F-irreducible, then P is *Harris recurrent* if for all $A \in \mathcal{B}(\mathsf{X})$ with $F(A) > 0$ and all $x \in \mathsf{X}$,

$$\Pr(\tau_A < \infty \mid X_0 = x) = 1.$$

Much of the following discussion on Harris recurrence for MCMC is based on an insightful paper by Gareth Roberts and Jeff Rosenthal [4]. For most results, alternative proofs based on the theory of harmonic functions are available [5].

Theorem 4.1. *The kernel P is Harris recurrent if and only if for all $A \in \mathcal{B}(\mathsf{X})$ and all $x \in \mathsf{X}$*

$$\Pr(X_n \in A \text{ infinitely often}(n) \mid X_0 = x) = 1.$$

Proof. It is obvious that if $\Pr(X_n \in A \text{ infinitely often}(n) \mid X_0 = x) = 1$, then $\Pr(\tau_A < \infty \mid X_0 = x) = 1$.

Now assume P is Harris recurrent and suppose by way of contradiction that

$$\Pr(X_n \in A \text{ infinitely often}(n) \mid X_0 = x) \neq 1.$$

Then there exists $A \in \mathcal{B}(\mathsf{X})$ such that $F(A) > 0$, some $x \in \mathsf{X}$, and some N such that

$$\Pr(X_n \notin A \text{ for all } n \geq N \mid X_0 = x) > 0,$$

which, by integrating over X_N, implies there exists $y \in \mathsf{X}$ such that

$$\Pr(\tau_A = \infty \mid X_0 = y) > 0.$$

This is a contradiction. □

Theorem 4.2. *If $FP = F$ and P is F-irreducible, then the following are equivalent.*

1. *P is Harris recurrent.*
2. *For all $x \in \mathsf{X}$, $A \in \mathcal{B}(\mathsf{X})$ with $F(A) = 1$, $\Pr(\tau_A < \infty \mid X_0 = x) = 1$.*
3. *For all $x \in \mathsf{X}$, $A \in \mathcal{B}(\mathsf{X})$ with $F(A) = 0$, $\Pr(X_n \in A \text{ for all } n \mid X_0 = x) = 0$.*

Proof. It is obvious that 1 implies 2 and that 2 holds if and only if 3 holds. That 3 implies 1 requires more work [4, Theorem 6]. □

Theorem 4.3. *Every F-irreducible Metropolis-Hastings Markov chain is Harris recurrent.*

Proof. Metropolis-Hastings kernels can be written as

$$P(x, A) = [1 - s(x)]M(x, A) + s(x)\delta_x(A)$$

with δ_X a point mass at x, $M(x, \cdot)$ is the kernel conditional on $X_{n+1} \neq X_n$, and

$$s(x) = \int q(x, y)[1 - \alpha(x, y)]\mu(\mathrm{d}y).$$

Notice that $M(x, \cdot)$ is absolutely continuous with respect to μ.

Since P is F-irreducible and $F(\{x\}) < 1$ it follws that $s(x) < 1$ for all x. Suppose $F(A) = 1$. Then $F(A^c) = 0$ and since $f(x) > 0$ on X, $\mu(A^c) = 0$. Hence, by absolute continuity, $M(x, A^c) = 0$, which implies $M(x, A) = 1$. Thus, if the current state is x, the chain will eventually move according to $M(x, \cdot)$ and it will necesarily move into A. Hence $\Pr(\tau_A < \infty \mid X_0 = x) = 1$. □

Consider componentwise samplers. Assume that $q_i(y_i \mid x)$ and that

$$q_i(y_i \mid x_1, \ldots, x_{i-1}, z_i, x_{i+1}, \ldots, X_d) > 0$$

if and only if

$$q_i(z_i \mid x_1, \ldots, x_{i-1}, y_i, x_{i+1}, \ldots, X_d) > 0.$$

This assumption is primarily useful for avoiding technicalities.

Lemma 4.4. *Let D_n be the event that there has not been a move in every coordinate direction. If $F(A) = 0$, then*

$$P(X_n \in A \mid X_0 = x) \leq P(D_n \mid X_0).$$

Theorem 4.5. *Suppose P is a conditional Metropolis-Hastings kernel. If $FP = F$, P is F-irreducible, and for all $X_0 = x$, with probability 1 there will eventually be a move in at least once in every coordinate direction, then P is Harris recurrent.*

Proof. The hypotheses imply that for all $x \in \mathsf{X}$

$$\lim_{n\to\infty} P(D_n \mid X_0 = x) = 0.$$

Suppose $F(A) = 0$. Since $f(x) > 0$ on X it follows that $\mu(A) = 0$. By the lemma,

$$P(X_n \in A \mid X_0 = x) \leq \lim_{n\to\infty} P(X_n \in A \mid X_0 = x) \leq \lim_{n\to\infty} P(D_n \mid X_0 = x) = 0.$$

The result follows from part 3 of Theorem 4.2. □

Corollary 4.1. *Every F-irreducible Gibbs sampler is Harris recurrent.*

5 Minorization

Definition 5.1. A *minorization condition* holds if there exists a positive integer m, a constant $\epsilon > 0$, and a probablity measure Q such that for all $x \in C \subseteq \mathsf{X}$ and all $A \in \mathcal{B}(\mathsf{X})$

$$P^m(x, A) \geq \epsilon Q(A). \tag{2}$$

The set C is said to be *small*.

It is a deep result that small sets are guaranteed to exist [3]. The focus here will mostly be on minorization conditions with $m = 1$ since this will often suffice in MCMC settings and the more general case is similar, but requires a lot more bookkeeping. The next few examples introduce some basic techniques for finding minorization conditions.

Example 5.1. Consider an independence sampler with proposal density $q(x)$ and suppose that

$$\epsilon = \inf_{x \in \mathsf{X}} \frac{q(x)}{f(x)} > 0.$$

Then

$$\begin{aligned} P(x, A) &\geq \int_A q(y) \left(1 \wedge \frac{f(y)q(x)}{f(x)q(y)}\right) \mathrm{d}y \\ &= \int_A \left(q(y) \wedge \frac{f(y)q(x)}{f(x)}\right) \mathrm{d}y \\ &= \int_A f(y) \left(\frac{q(y)}{f(y)} \wedge \frac{q(x)}{f(x)}\right) \mathrm{d}y \\ &\geq \epsilon F(A) \end{aligned}$$

and hence X is small.

Example 5.2. Fix a point $x^* \in (0, 1)$ and consider the so-called Beta walk

$$\begin{aligned} k(y \mid x) &= \frac{1}{2}\frac{1}{x} I(0 < y < x) + \frac{1}{2}\frac{1}{1-x} I(x < y < 1) \\ &\geq \frac{1}{2}\frac{1}{x} I(0 < y < x) \\ &\geq \frac{1}{2}\frac{1}{x} I(0 < y < x) I(0 < y < x^*) I(x > x^*) \\ &\geq \frac{1}{2}\frac{1}{x} I(0 < y < x^*) I(x > x^*). \end{aligned}$$

Choose $C = [a, b] \subseteq (0, 1)$ so that $a < x^* < b$. Then, for all $x \in \mathsf{X}$,

$$k(y \mid x) \geq \frac{1}{x^*} I(0 < y < x^*) \frac{x^*}{2b}.$$

Hence, C is small with $\epsilon = x^*/2b$ and $Q = \text{Uniform}(0, x^*)$.

Example 5.3. Consider a two-variable deterministic scan Gibbs sampler. Recall that the MTF is

$$k_{DG}(x', y \mid x, y) = f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x').$$

Let C be such that

$$\inf_{y \in C} f_{X|Y}(x' \mid y) = h(x') > 0$$

and set

$$\epsilon = \int h(x) f_{Y|X}(y \mid x) \mathrm{d}x \mathrm{d}y.$$

Then, for all $y \in C$,

$$k_{DG}(x', y' \mid x, y) \geq \epsilon \, \frac{h(x') f_{Y|X}(y' \mid x')}{\epsilon} = \epsilon q(x', y')$$

and hence C is small.

Example 5.4. Consider a two-variable random scan Gibbs sampler with selection probability r and note that

$$P_{RG}^2((x, y), A) = \int P_{RG}((x, y), \mathrm{d}(u, v)) P_{RG}((u, v), A) \geq r(1 - r) P_{DG}((x, y), A).$$

To obtain a minorization condition for random scan Gibbs samplers, it suffices to find a minorization condition for deterministic scan Gibbs samplers.

Minorization is fundamental to much of modern Markov chain theory because, at least in part, it allows the construction of the so-called *split chain* [2]. Suppose a set C is small with $m = 1$. When $x \in C$ notice that

$$R(x, A) := \frac{P(x, A) - \epsilon Q(A)}{1 - \epsilon}$$

is a probability measure and is called the *residual measure*. Now, a mixture representation of the kernel P is available

$$P(x, A) = \epsilon Q(A) + (1 - \epsilon) R(x, A).$$

This suggests an algorithm for simulating the split chain, which is an augmented Markov chain on $\mathsf{X} \times \{0, 1\}$ having a marginal given by the kernel P.

Algorithm 1 Split Chain

Input: Current value $X_n = x$.
if $x \notin C$ **then**
    draw $X_{n+1} \sim P(x, \cdot)$
else if $x \in C$ **then**
    independently draw $\delta \sim \text{Bernoulli}(\epsilon)$
    if $\delta = 1$ **then**
        draw $X_{n+1} \sim Q(\cdot)$
    else if $\delta = 0$ **then**
        draw $X_{n+1} \sim R(x, \cdot)$
    end if
end if

Notice that every time $X_{n+1} \sim Q(\cdot)$, the Markov chain "resets." Later this construction will be used to break the Markov chain into independent "tours."

While Algorithm 1 works is an extraordinarily useful theoretical construction, in practice it is not at all useful. The primary difficulty is that simulation from the residual measure, $R(x, \cdot)$, is impractical in practically relevant settings.

6 Coupling and Convergence

The idea is to construct two Markov chains that, marginally, are copies, but will eventually become equal, that is, couple.

Suppose C is small with $m = 1$. Initialize one copy from an arbitrary distribution and one from the target distribution, that is, $X_0 \sim \Lambda$ and $X'_0 \sim F$. Then, given X_n and X'_n,

1. If $X_n = X_n'$, let $X_{n+1} = X_{n+1}' \sim P(X_n, \cdot)$.

2. If $X_n \neq X_n'$:

   (a) If $(X_n, X_n') \in C \times C$, then

      i. with probability ϵ, $X_{n+1} = X_{n+1}' \sim Q(\cdot)$

      ii. with probability $1-\epsilon$, $X_{n+1} \sim R(X_n, \cdot)$ and $X_{n+1}' \sim R(X_n', \cdot)$

   (b) If $(X_n, X_n') \notin C \times C$, then $X_{n+1} \sim P(X_n, \cdot)$ and $X_{n+1}', \sim P(X_n', \cdot)$

It is left as an exercise to verify that

$$\Pr(X_n \in A) \mid X_0 = x) = P^n(x, A)$$

and that

$$\Pr(X_n' \in A \mid X_0' = x) = F(A).$$

Let T be the (random) time at which coupling occurs, that is, $T = \inf\{n \geq 1 \mid \delta_n = 1\}$.

Lemma 6.1 ([1])**.** *If the Markov chain is aperiodic and Harris recurrent, then $T < \infty$ and, for all x, as $n \to \infty$, $\Pr_x(T > n) \to 0$.*

Lemma 6.2. *For all x,*

$$\|P^n(x, \cdot) - F(\cdot)\| \leq \Pr_x(T > n).$$

Proof. From the split chain construction:

$$\begin{aligned}
|P^n(x, A) - F(A)| &= |\Pr(X_n \in A) - \Pr(X_n' \in A)| \\
&= |\Pr(X_n \in A, X_n = X_n') + \Pr(X_n \in A, X_n \neq X_n') \\
&\quad - \Pr(X_n' \in A, X_n = X_n') - \Pr(X_n' \in A, X_n \neq X_n')| \\
&= |\Pr(X_n \in A, X_n \neq X_n') - \Pr(X_n' \in A, X_n \neq X_n')| \\
&\leq \Pr(X_n \in A, X_n \neq X_n') \vee \Pr(X_n' \in A, X_n \neq X_n') \\
&\leq \Pr(X_n \neq X_n') \\
&\leq \Pr(T > n).
\end{aligned}$$

$\square$

Combining Lemmas 6.1 and 6.2 yields the following result.

Theorem 6.3. *If the kernel P satisfies $FP = F$ and is F-irreducible, aperiodic, and Harris recurrent, then, for all x,*

$$\lim_{n\to\infty} \|P^n(x,\cdot) - F(\cdot)\| = 0.$$

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